Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic decisions

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Fields

Customer, purchase, and behavior metrics

37

Missing Values

Review ratings imputed

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Python Data Preparation

Data Cleaning

- Imported with pandas
- Checked structure and statistics
- Handled missing review ratings
- Standardized column names

Feature Engineering

- Created age groups
- Calculated purchase frequency
- Dropped redundant promo codes
- Loaded to PostgreSQL

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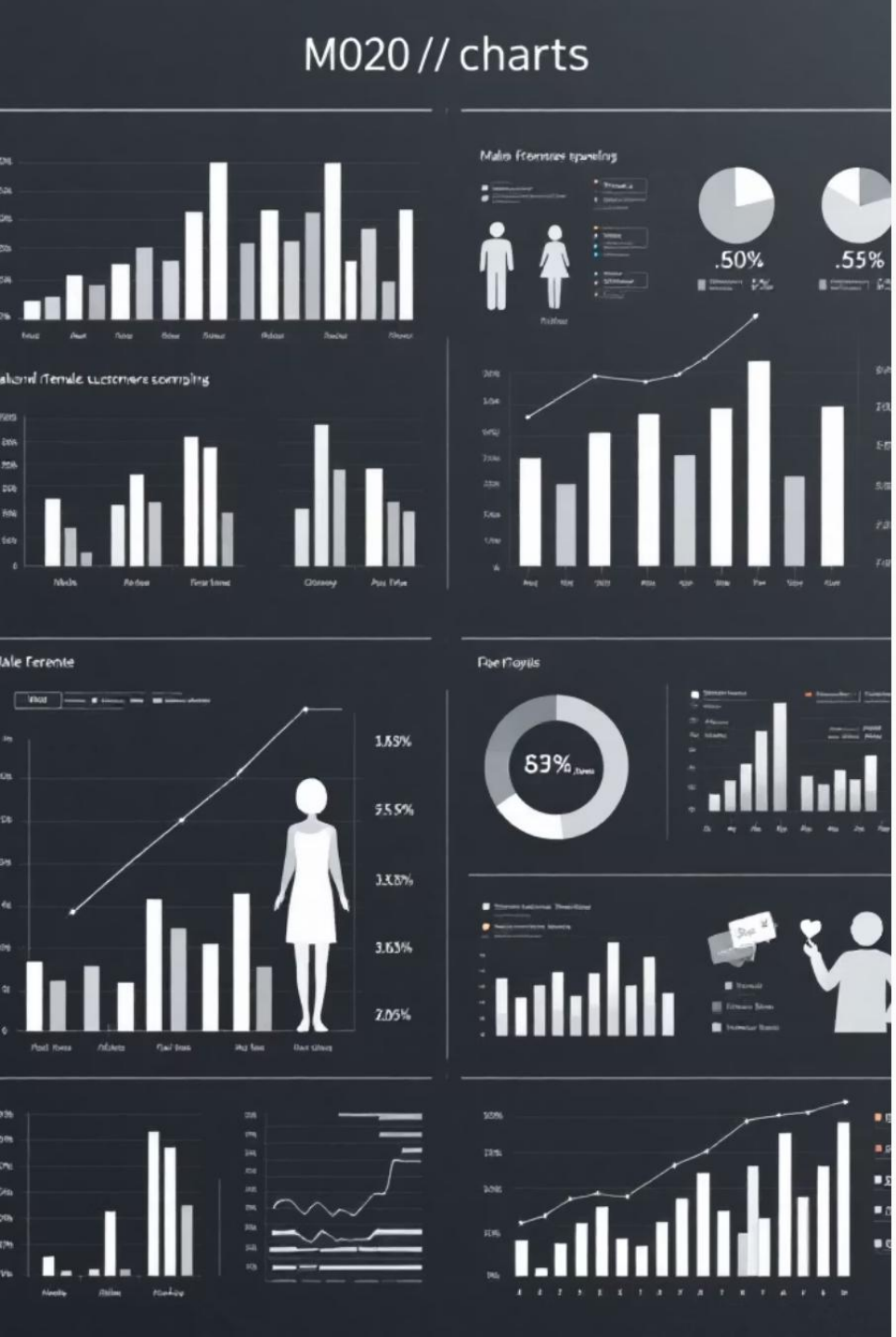

SQL Analysis: Revenue Insights

Gender Revenue

Female: $75,191

Male: $157,890

Shipping Comparison

Standard: $58.46

Express: $60.48

Discount Users

839 high-spending customers used discounts

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Product Performance

1

Top-Rated Products

Gloves (3.86), Sandals (3.84), Boots (3.82)

2

Discount-Heavy Items

Hat (50%), Sneakers (49.66%), Coat (49.07%)

3

Best-Selling by Category

Clothing: Blouse & Pants (171), Accessories: Jewelry (171)

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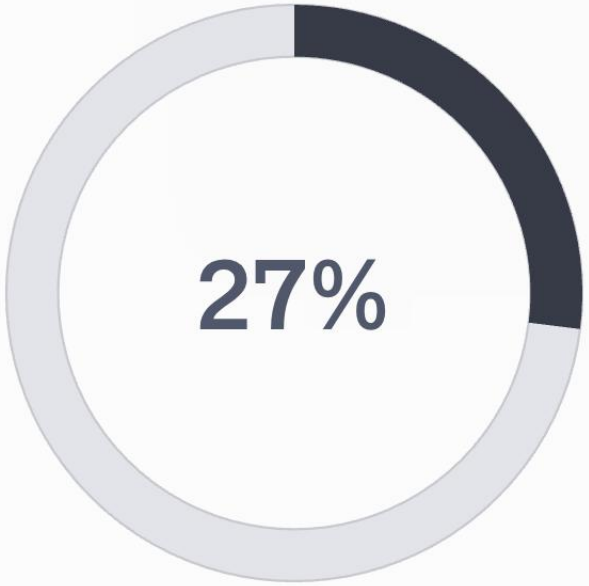
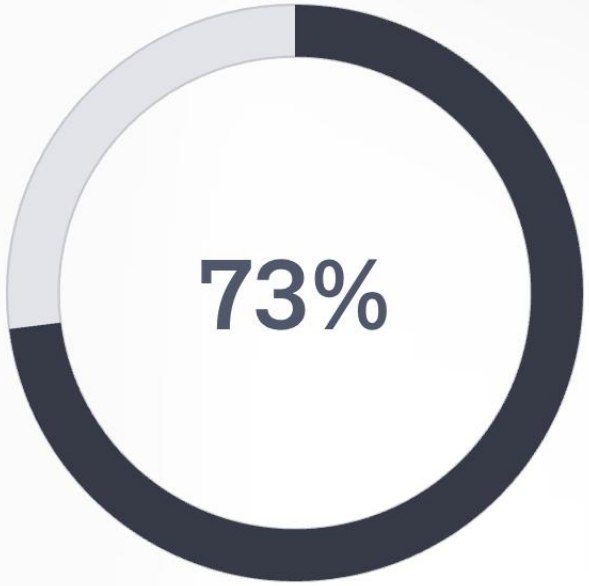
Subscription Analysis



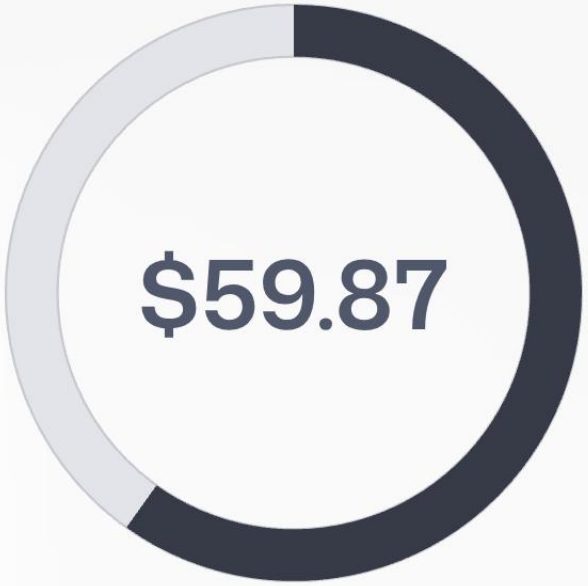
Subscribers

1,053 customers



Non-Subscribers

2,847 customers



Avg Spend

Non-subscribers

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Customer Segmentation

Loyal

3,116 customers with consistent purchases

Returning

701 customers making multiple visits

New

83 first-time buyers

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Age Group Performance

28%

Young Adult

$62,143 revenue

25%

Middle-aged

$59,197 revenue

24%

Adult

$55,978 revenue

23%

Senior

$55,763 revenue

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Power BI Dashboard

01

KPIs Displayed

3.9K customers, $59.76 avg purchase, 3.75 avg rating

02

Visualizations

Revenue by category, sales by age group, subscription status

03

Interactive Filters

Gender, category, shipping type, subscription status

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits

Customer Loyalty

Reward repeat buyers

Review Discount Policy

Balance sales with margins

Targeted Marketing

Focus on high-revenue segments

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